GENERAL SENSORY PATHWAY

=An Excerpt from Prof. Yama O.E.'s Teaching Notes Series for Medical Students=

Outline:

1. **OVERVIEW & CONCEPTUAL FRAMEWORK**
2. **FUNDAMENTAL COMPONENTS OF ANY SENSORY PATHWAY**
3. **PATHWAYS FOR GENERAL SENSATIONS FROM THE BODY (Below the Head)**
4. **PATHWAYS FOR GENERAL SENSATIONS FROM THE HEAD**
5. **SUMMARY COMPARISON TABLE**
6. **CLINICAL CORRELATES — LESIONS OF SENSORY PATHWAYS**
7. **QUICK MEMORY AID — Pathway Summary in Plain Language**

1. OVERVIEW & CONCEPTUAL FRAMEWORK

The nervous system does 2 fundamental things:

a) it **receives information** from the world (sensory/afferent function)

b) and **sends commands** to execute responses (motor/efferent function). Sensory pathways — also called **ascending pathways** — carry impulses upward from peripheral receptors to the cerebral cortex, where they are consciously perceived.

General sensations are those felt everywhere in the body — skin, muscles, joints, and viscera. They include **touch, pain, temperature, pressure,** and **proprioception**. This is in contrast to **special senses** (vision, hearing, smell, taste, and balance), which have dedicated cranial organs. This write-up covers only general sensations.

> 📌 *Think of sensory pathways like a postal relay system: a letter (stimulus) is picked up by a courier (1st-order neuron), taken to a regional hub (2nd-order neuron that crosses to the other side), then to a national hub (3rd-order neuron in the thalamus), and finally delivered to the destination (sensory cortex).*

2. FUNDAMENTAL COMPONENTS OF ANY SENSORY PATHWAY

Every **general sensory pathway** regardless of the modality it carries is built from the same 4 elements:

(*i*). Receptors

Receptors are specialized sensory organs (transducers) located in the skin, joints, bones, muscles, and viscera. They convert a physical or chemical stimulus (touch, heat, stretch) into an electrical nerve impulse. The type of receptor determines the type of sensation processed:

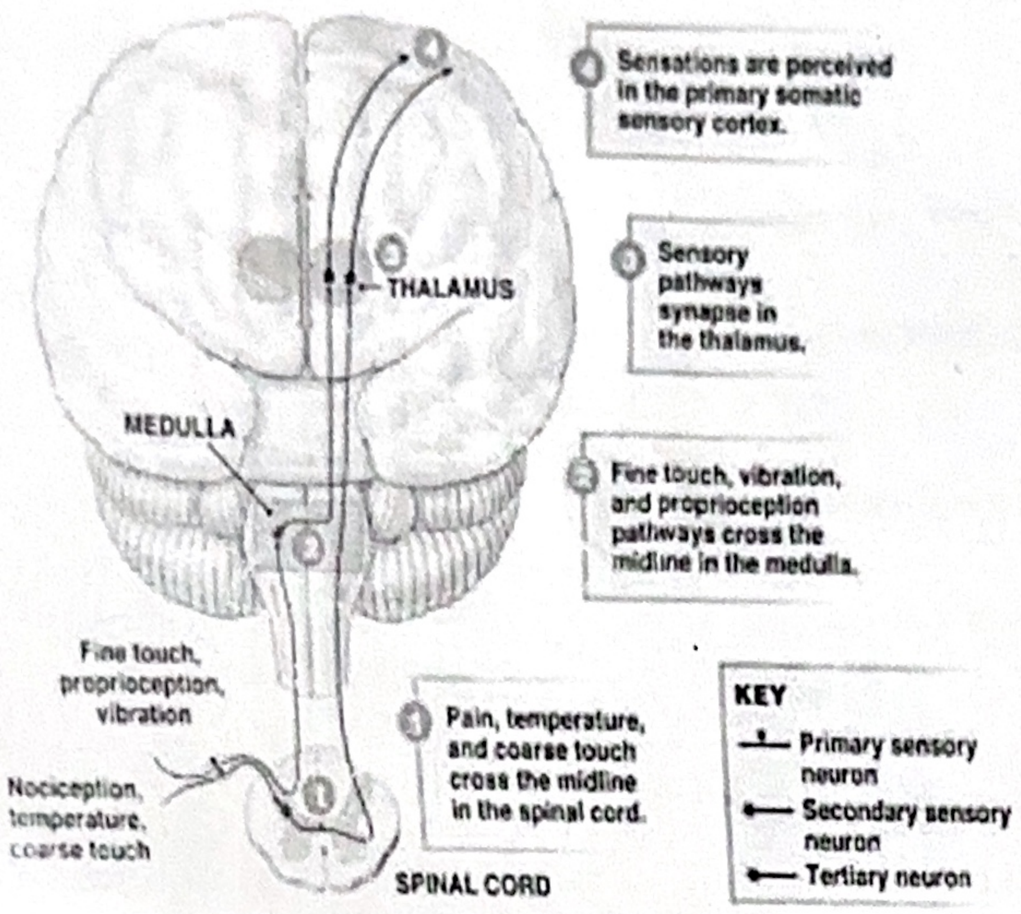
- **Meissner's corpuscles** — light discriminative touch (fingertips, palms)
- **Merkel's discs** — sustained pressure and fine spatial detail
- **Ruffini endings** — deep skin tension and stretch
- **Pacinian corpuscles** — vibration and deep pressure
- **Muscle spindles & Golgi tendon organs** — proprioception (body position and muscle tension)
- **Free nerve endings** — pain (nociception) and temperature

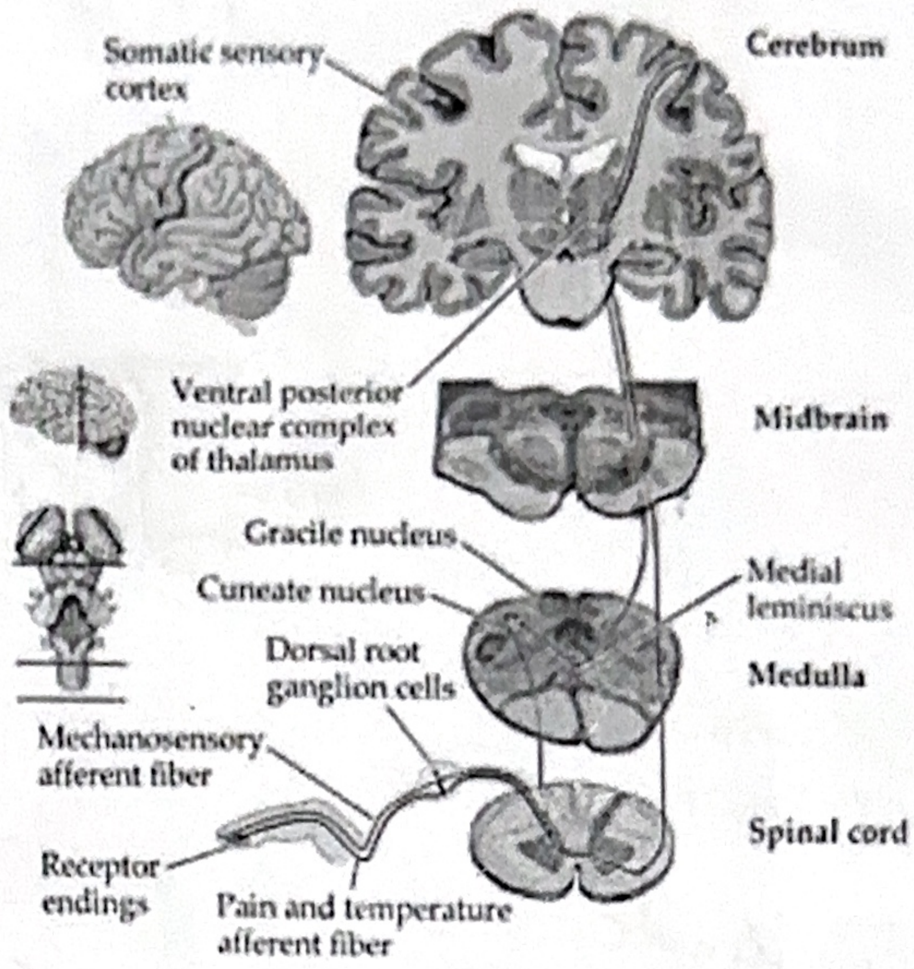
(*ii*). First-Order Neurons

First-order neurons are the primary sensory neurons. Their cell bodies sit in ganglia **outside** the central nervous system (CNS) — specifically in the **dorsal root (spinal) ganglia** for the body, and in the **trigeminal ganglion** for the head. They are typically **pseudo-unipolar cells**: a single process that *splits*

into a peripheral branch (receiving from receptors) and a central branch (axon entering the CNS to synapse on 2nd-order neurons).

> 📌 *The exception: proprioceptive first-order neurons for the HEAD have their cell bodies inside the CNS itself, in the mesencephalic nucleus of the trigeminal nerve — the only sensory ganglion cell located within the brain.*





(*iii*). Second-Order Neurons

Second-order neurons reside inside the CNS. Their most critical feature is that their axons **cross the midline (decussate)**. This is why a lesion on one side of the brain causes sensory loss on the **opposite (contralateral)** side of the body. After crossing, their axons ascend in organized **tracts** (bundles of nerve fibers with the same function) within the white matter.

> 📌 *Important: WHEN the decussation occurs differs between pathways. Posterior column (fine touch) decussates at the MEDULLA; spinothalamic (pain/temp) decussates at the SPINAL CORD LEVEL. This distinction is the key to understanding Brown-Séquard syndrome.*

(*vi*). Third-Order Neurons

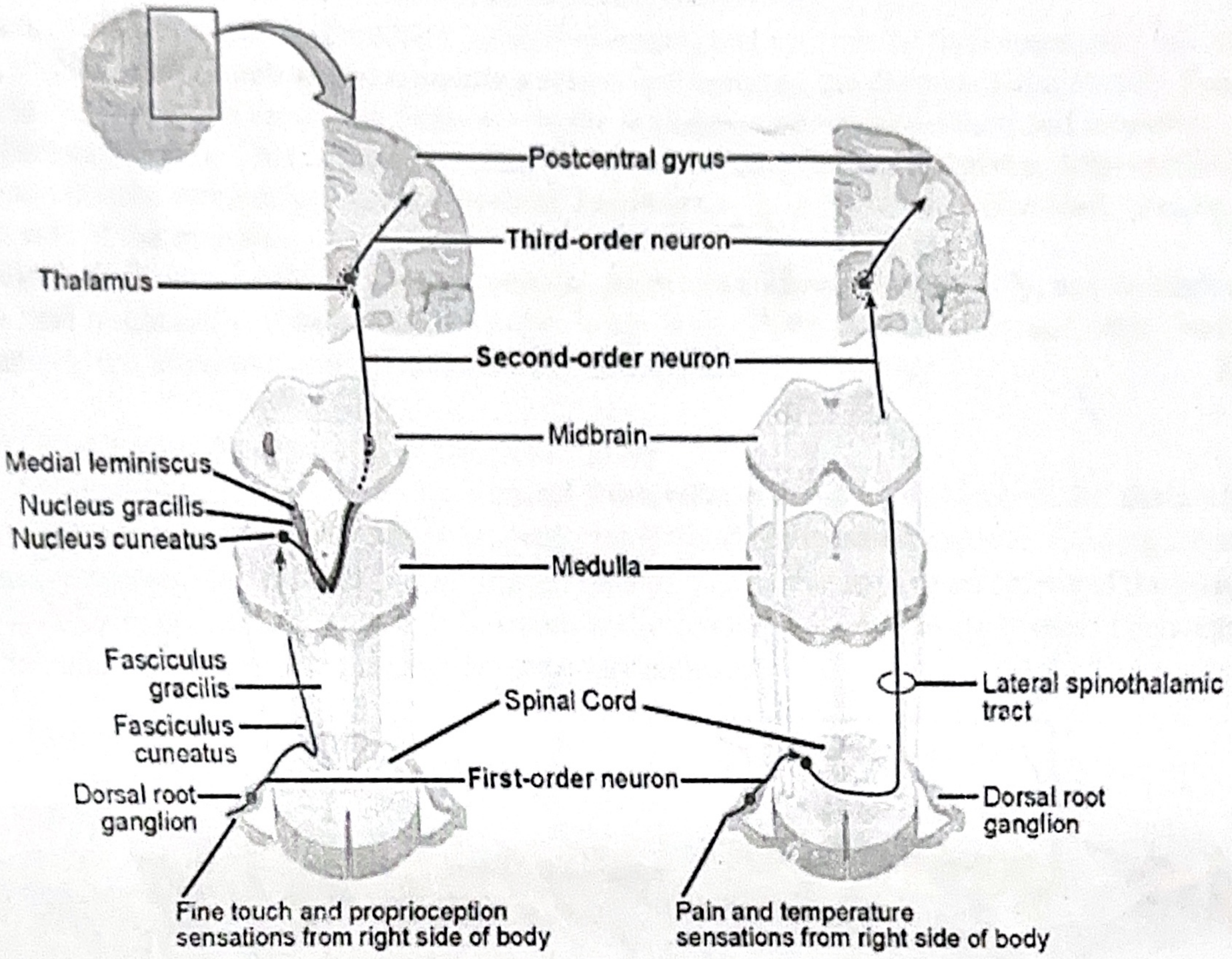
Third-order neurons are located in the **thalamus** — the brain's grand relay station. For general **body** sensations, they sit in the **Ventral Posterolateral (VPL) nucleus**; for **head** sensations, in the **Ventral Posteromedial (VPM) nucleus**. Their axons project through the posterior limb of the internal capsule (as **thalamic radiations**) to reach the **primary somatosensory cortex** in the postcentral gyrus of the parietal lobe (Brodmann areas 3, 1, and 2).

3. PATHWAYS FOR GENERAL SENSATIONS FROM THE BODY (Below the Head)

Below the level of the head, general sensory pathways are organized into **2 major systems** that run in parallel:

i) the ***Dorsal Column–Medial Lemniscus system*** and
ii) the ***Spinothalamic (Anterolateral) system.***

These two highways carry fundamentally different types of information.



(i). Dorsal Column – Medial Lemniscus (DCML) Pathway

Modalities carried: Fine (discriminative) touch, vibration sense, conscious proprioception (joint position sense), and 2-point discrimination.

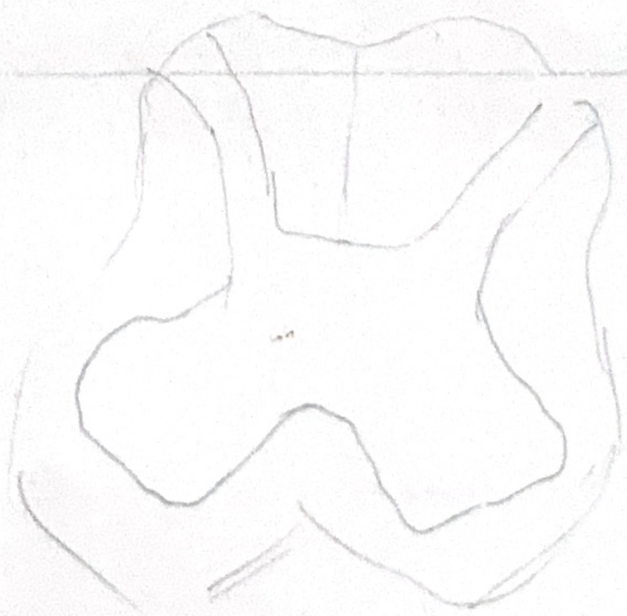
This is the 'precision pathway' it carries detailed spatial and quality information so you can recognize an object just by touch, feel a vibrating tuning fork, and know exactly where your limbs are without looking.

Step 1: First-Order Neuron → Spinal Cord

Cell bodies sit in the dorsal root (spinal) ganglia. Their peripheral processes receive signals from receptors in skin and joints. Their central processes (axons) enter the spinal cord via the dorsal root and — uniquely — **do NOT synapse immediately**. Instead, they ascend directly in the **ipsilateral (same-side) posterior column (funiculus)** of white matter, all the way to the medulla.

Within the posterior column, the axons are organized into two named tracts:

i) ***Fasciculus Gracilis (Gracile tract)****: runs more medially; carries signals from the lower limb and lower trunk (below T6)*



ii) ***Fasciculus Cuneatus* (*Cuneate tract*)**: *runs laterally to gracilis; carries signals from the upper limb and upper trunk (above T6)*

Memory tip: 'Gracilis' means slender/thin — it carries signals from the LOWER body. The Cuneate tract (named after the cuneate nucleus it ends in) carries signals from the UPPER body and enters the cord only at thoracic levels and above.

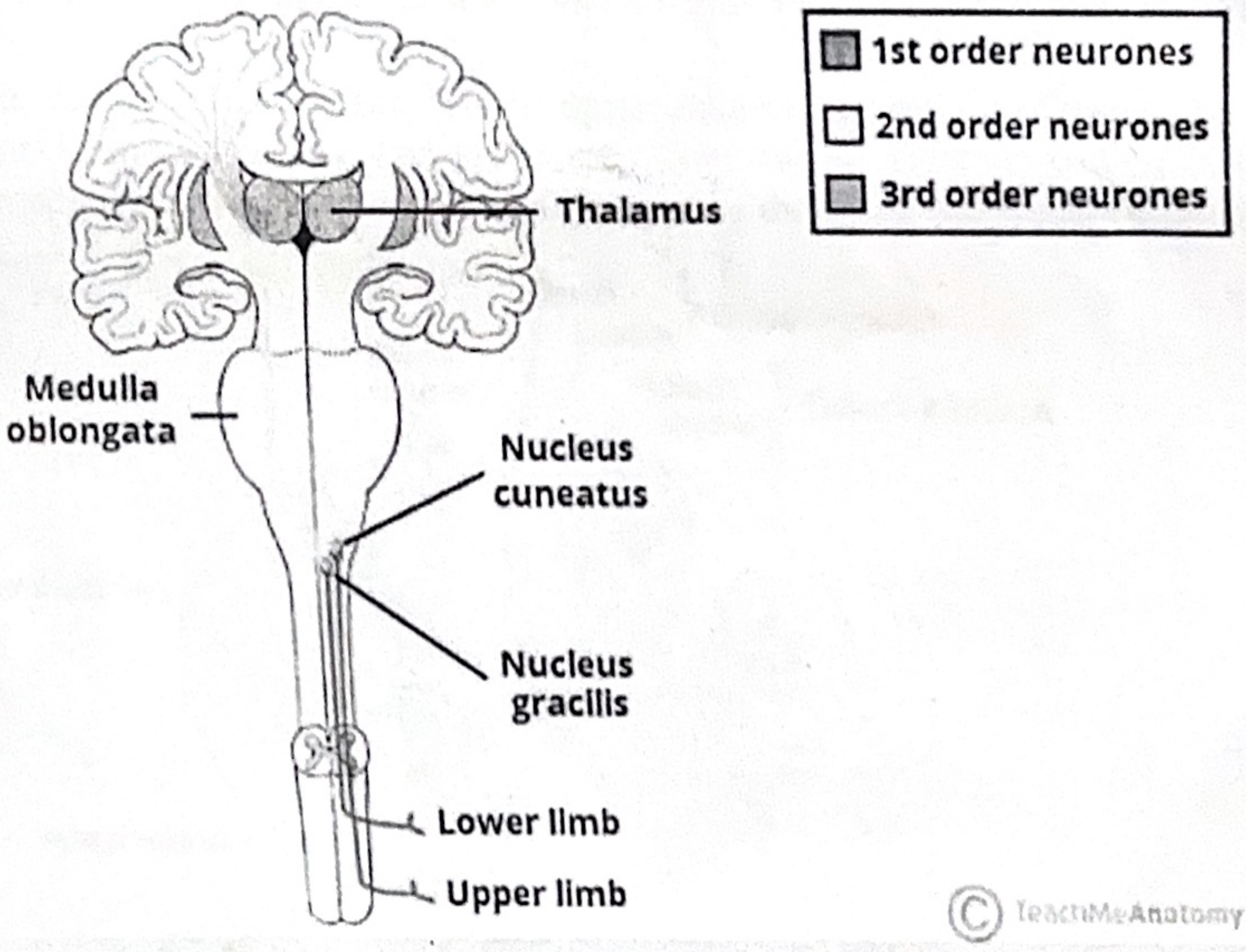
Step 2: Second-Order Neuron → Decussation in the Medulla

Both tracts ascend to the caudal medulla (lower brainstem) and synapse on their respective second-order neurons in the **Gracile nucleus** and **Cuneate nucleus** (collectively, the dorsal column nuclei). From here, the second-order neurons send their axons across the midline — sweeping forward and medially — as the **internal arcuate fibers**. This crossing is called the **lemniscal (sensory) decussation**. After crossing, these fibers immediately reorganize into the **Medial Lemniscus** — a flat ribbon-like tract running on the opposite side of the midline.

The medial lemniscus ascends through the medulla, pons, and midbrain. Importantly, as it ascends through the pons and midbrain, it **rotates 90° laterally**: lower body fibers shift laterally and upper body fibers shift medially (an important clinical concept for locating brainstem lesions).

Step 3: Third-Order Neuron → Thalamus to Cortex

The medial lemniscus terminates in the **Ventral Posterolateral (VPL) nucleus of the thalamus**. VPL neurons (3rd-order) send axons through the **posterior limb of the internal capsule**, forming the **superior thalamic radiation**. These fibers fan out (radiate) to terminate in the **upper two-thirds of the postcentral gyrus** — Brodmann areas 3, 1, and 2 — which is the primary somatosensory cortex. The somatotopic map (homunculus) is preserved throughout the entire pathway.



CLINICAL PEARL: A lesion in the POSTERIOR COLUMN causes IPSILATERAL loss of fine touch, vibration, and proprioception BELOW the lesion level. This is tested clinically with the Romberg test (patient becomes unsteady when eyes are closed, since they cannot rely on proprioception). Vitamin B12 deficiency (subacute combined degeneration) classically damages the posterior columns.

(ii). The Spinothalamic (Anterolateral) System

This system carries 2 distinct but related modalities that share most of their pathway:

- pain and temperature in one tract,
- crude touch and pressure in another tract.

Both cross in the spinal cord (not the brainstem), which is a key distinguishing feature.

(a). Lateral Spinothalamic Tract — Pain and Temperature

Modalities: Pain (nociception) and Temperature (thermoception).

Step 1: First-Order Neuron

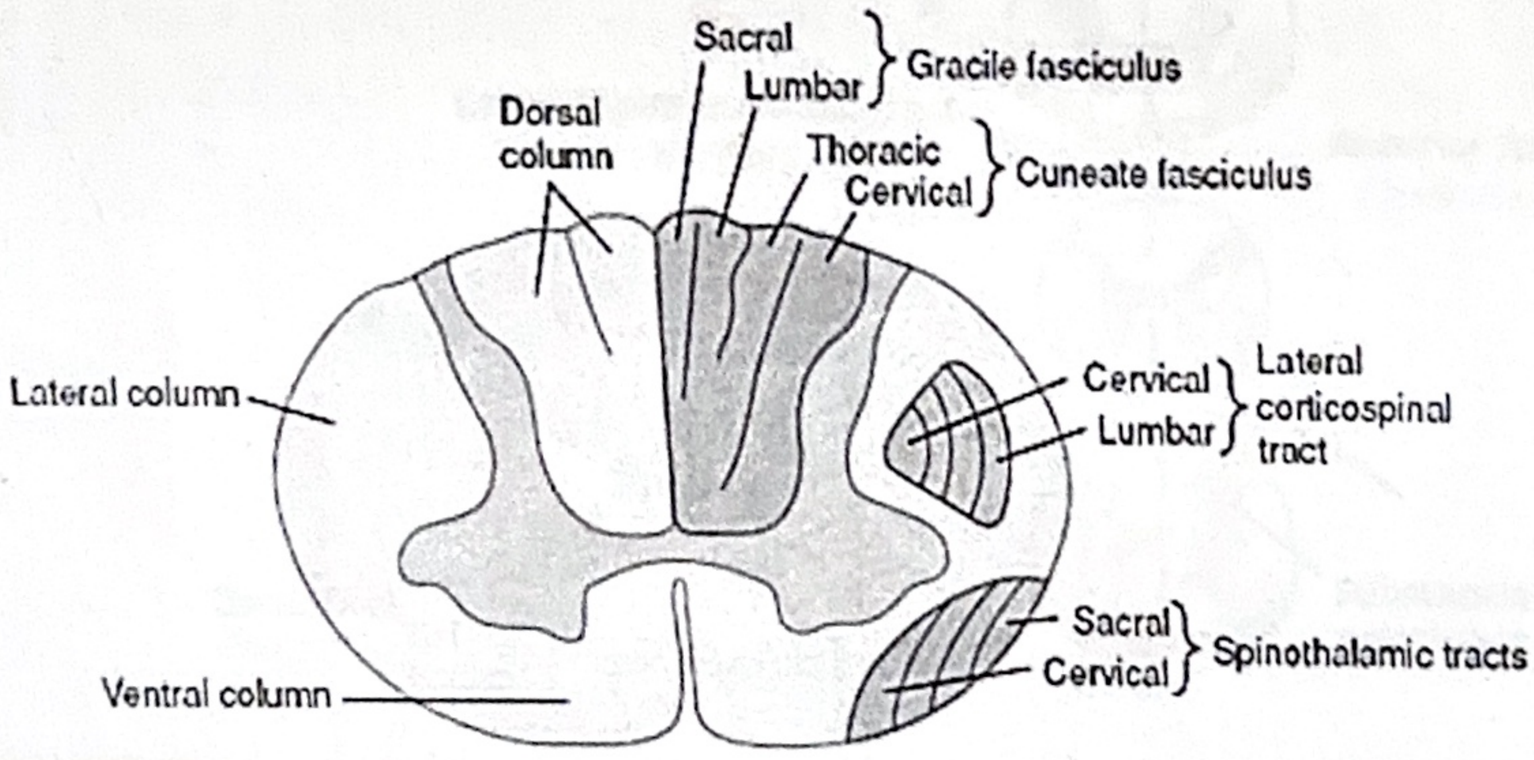
Cell bodies are in the dorsa! root ganglia. Pain fibers are carried by **small-diameter, thinly myelinated Aδ fibers** (sharp, fast pain) and **unmyelinated C fibers** (slow, dull, burning pain). Their central axons enter the spinal cord and ascend or descend **1–2 spinal segments** in the **dorsolateral tract of Lissauer** (also called the posterolateral fasciculus) before synapsing on second-order neurons in the **substantia gelatinosa** (Rexed's lamina II) of the posterior gray horn.

> 📌 *The tract of Lissauer is why pain can be felt 1–2 segments above or below the actual injury level — the first-order fibers travel a short distance before synapsing.*

Step 2: Second-Order Neuron — Immediate Decussation

Second-order neurons cross to the opposite side almost immediately — right at the same spinal cord level — through the **ventral white commissure**. This is the key ***difference from the posterior column, which crosses much later in the medulla***. After crossing, they ascend in the **lateral white funiculus** as the **lateral spinothalamic tract**. In the medulla, the lateral and anterior spinothalamic tracts merge into a single bundle called the **spinal lemniscus**, which continues through the pons and midbrain to reach the thalamus.

Somatotopic organization in the lateral spinothalamic tract: sacral fibers lie most laterally, and cervical fibers lie most medially. This is clinically important — a central cord lesion (e.g., syringomyelia) affects the medially located cervical fibers first, sparing the sacral fibers (sacral sparing).



Step 3: Third-Order Neuron

The spinal lemniscus terminates in the **VPL nucleus of the thalamus**. Third-order neurons project through the posterior limb of the internal capsule to areas **3**, **1**, and **2 of the primary somatosensory cortex** in the postcentral gyrus.

> 🩺 *CLINICAL PEARL: A lesion of the LATERAL spinothalamic tract causes CONTRALATERAL loss of pain and temperature, typically 1–2 segments BELOW the lesion (because the first-order*

fiber travels 1–2 segments in Lissauer's tract before crossing). This is seen in Brown-Séquard syndrome.

(b). Anterior Spinothalamic Tract — Crude Touch and Pressure

Modalities: Crude (non-discriminative) touch and pressure — you know something touched you, but not exactly where or what it is.

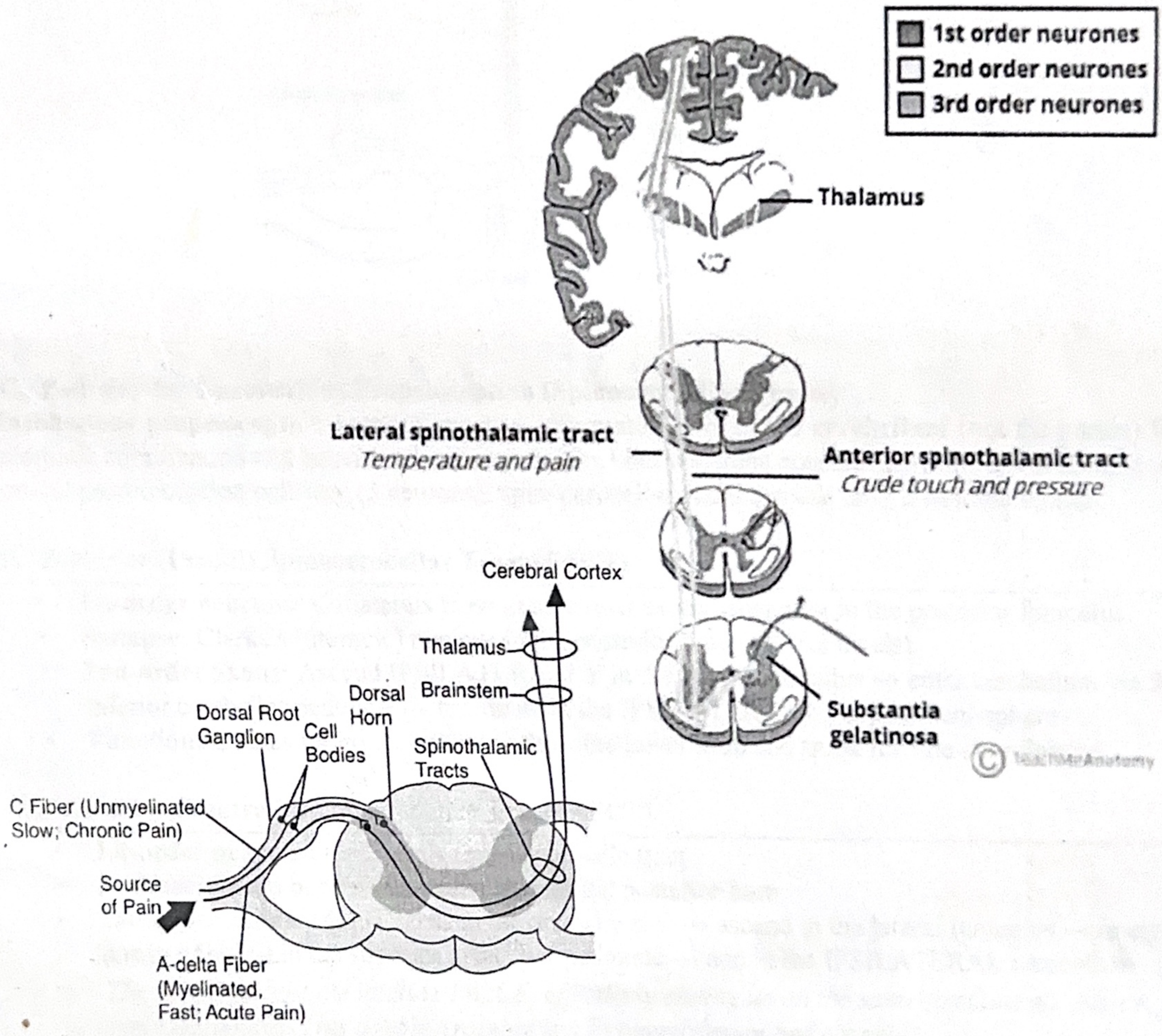
Steps 1–3 (same architecture as lateral spinothalamic, different destination in the cord):

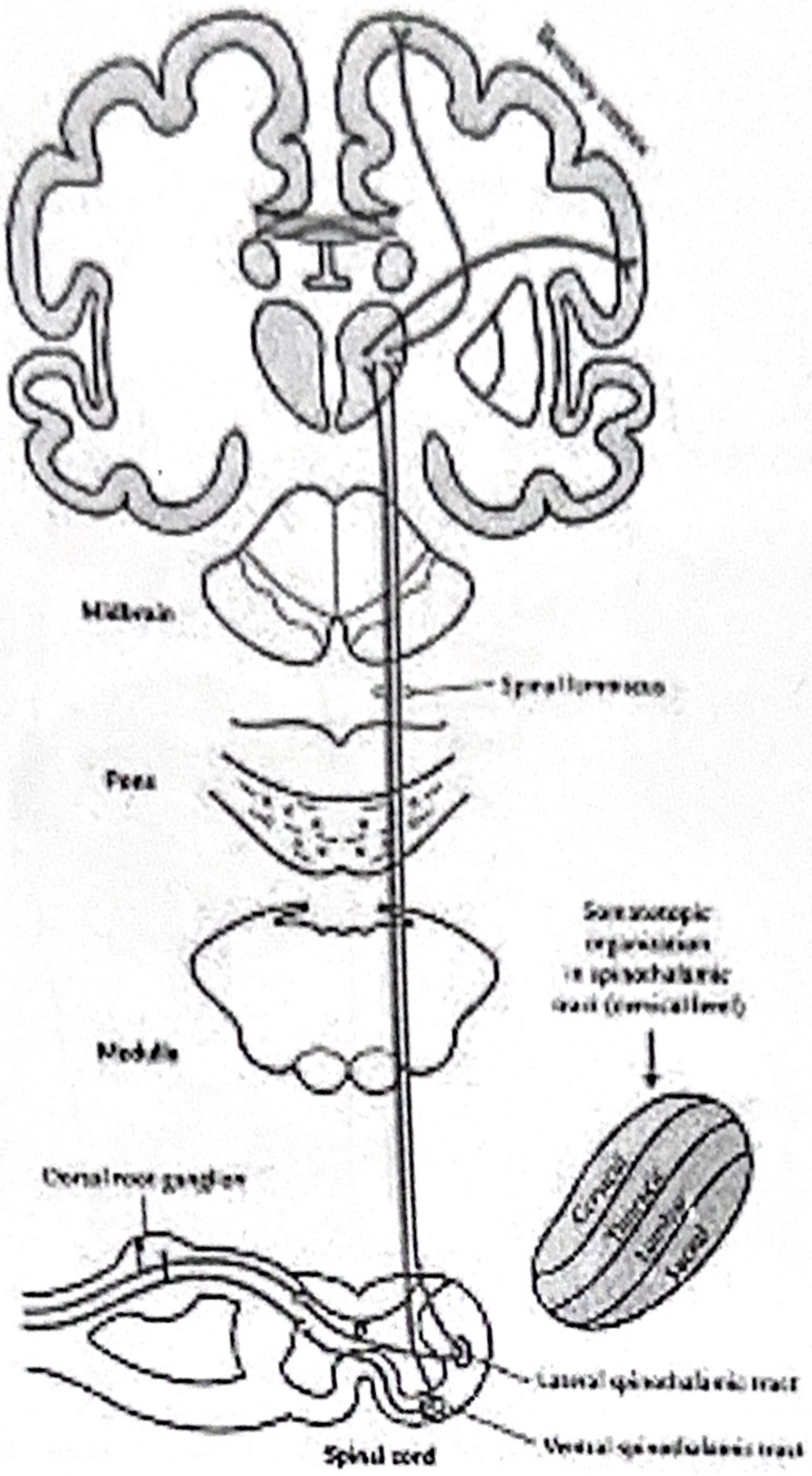
First-order neurons in the dorsal root ganglia send axons into the spinal cord, ascending 1–2 segments in the **dorsolateral tract of Lissauer**, then synapsing on second-order neurons in the **Nucleus Proprius** of the posterior gray horn.

Second-order neurons cross via the **ventral white commissure** and ascend in the **anterior (ventral) white funiculus** as the **anterior (ventral) spinothalamic tract**. In the medulla, this merges with the lateral spinothalamic tract to form the **spinal lemniscus,** which terminates in the **VPL nucleus of the thalamus.**

Third-order neurons project to **areas 3, 1,** and **2 of the sensory cortex.**

CLINICAL PEARL: Crude touch is partially preserved in posterior column lesions (because it also has some anterior spinothalamic tract representation), but fine discriminative touch is LOST. This is important diagnostically.





EuPrisma concentrate Mandile Grace

3C. Pathway for Unconscious Proprioception (Spinocerebellar Tracts)

Unconscious proprioception is proprioceptive information sent to the **cerebellum** (not the cortex) for automatic coordination and balance adjustments — the kind you don't consciously think about. Unlike the cortical proprioception pathway (3 neurons), spinocerebellar pathways use only **2-neuron chains**.

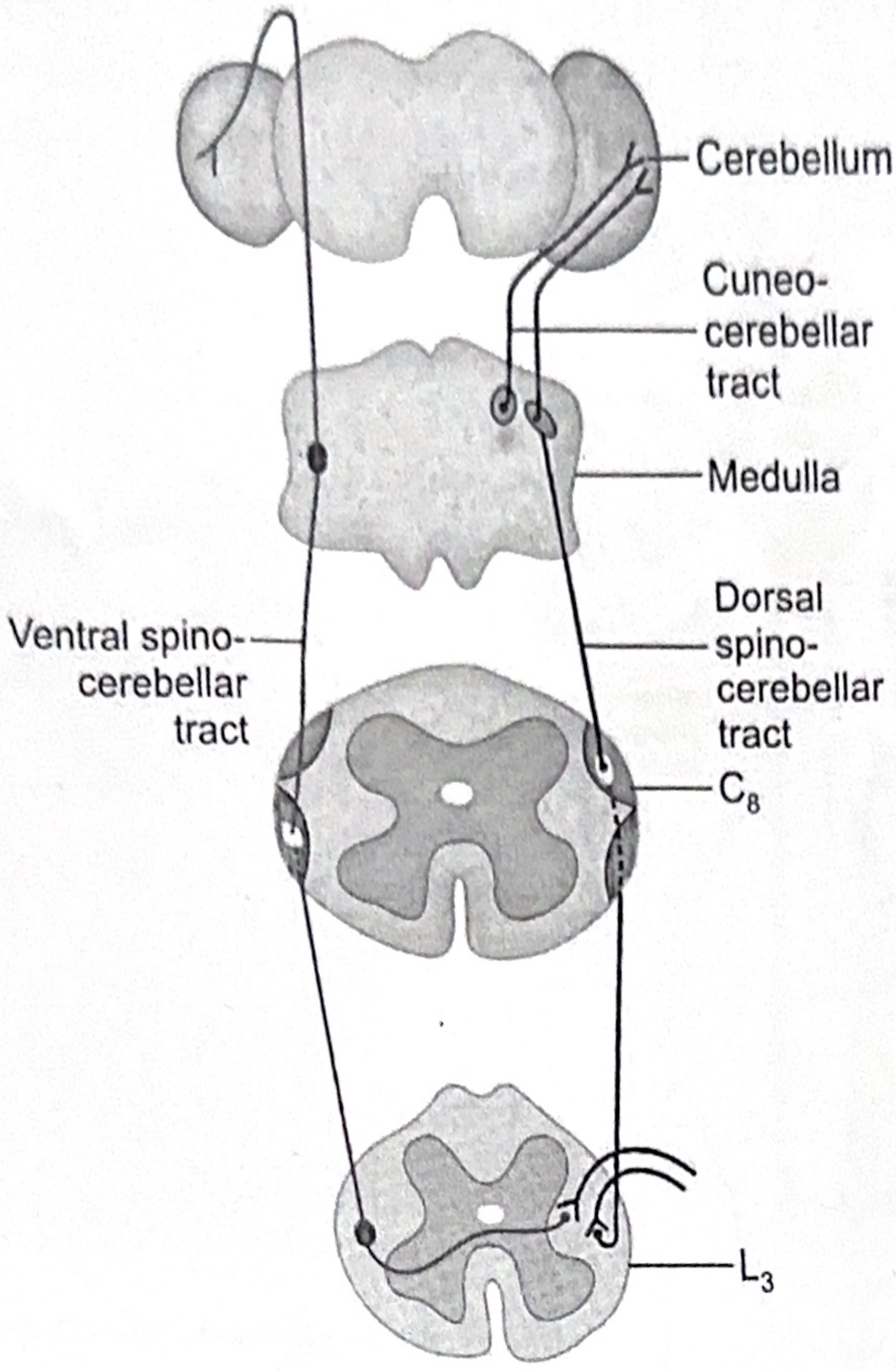
(*i*). Posterior (Dorsal) Spinocerebellar Tract (DSCT)

- **1st-order neurons:** Collaterals from gracile tract axons ascending in the posterior funiculus
- **Synapse:** Clarke's (thoracic) nucleus in the posterior horn (C8–L2 levels)
- **2nd-order axons:** Ascend IPSILATERALLY in the lateral funiculus → enter cerebellum via the inferior cerebellar peduncle → terminate in the IPSILATERAL cerebellar hemisphere
- **Function:** Carries proprioceptive info from the lower limb and trunk for fine coordination

(*ii*). Anterior (Ventral) Spinocerebellar Tract (ASCT)

- **1st-order neurons:** Collaterals from the gracile tract
- **Synapse:** Spinal border cells at the base of the posterior horn
- **2nd-order axons:** MOST cross to the opposite side → ascend in the lateral funiculus → re-cross (cross AGAIN) in the superior cerebellar peduncle → end in the IPSILATERAL cerebellum

> 📌 *The ASCT crosses the midline TWICE, effectively ending up on the same (ipsilateral) side as the original stimulus. This double-crossing is a frequently examined concept.*



(*iii*). Cuneocerebellar Tract

- Equivalent of the DSCT but for the UPPER limb and upper trunk (above T6)
- **2nd-order neurons:** Accessory cuneate nucleus in the rostral medulla
- Axons travel as posterior external arcuate fibers → inferior cerebellar peduncle → ipsilateral cerebellar hemisphere

(*vi*). Vestibulocerebellar Tract

- Carries signals from the vestibule and semicircular canals (balance organs) in the inner ear
- **1st-order neurons:** Vestibular ganglion → vestibular division of CN VIII
- **DIRECT fibers:** Ipsilateral inferior cerebellar peduncle → cerebellar hemisphere (no synapse at vestibular nuclei)
- **INDIRECT fibers:** Synapse in vestibular nuclei first → then inferior cerebellar peduncle → cerebellar hemisphere

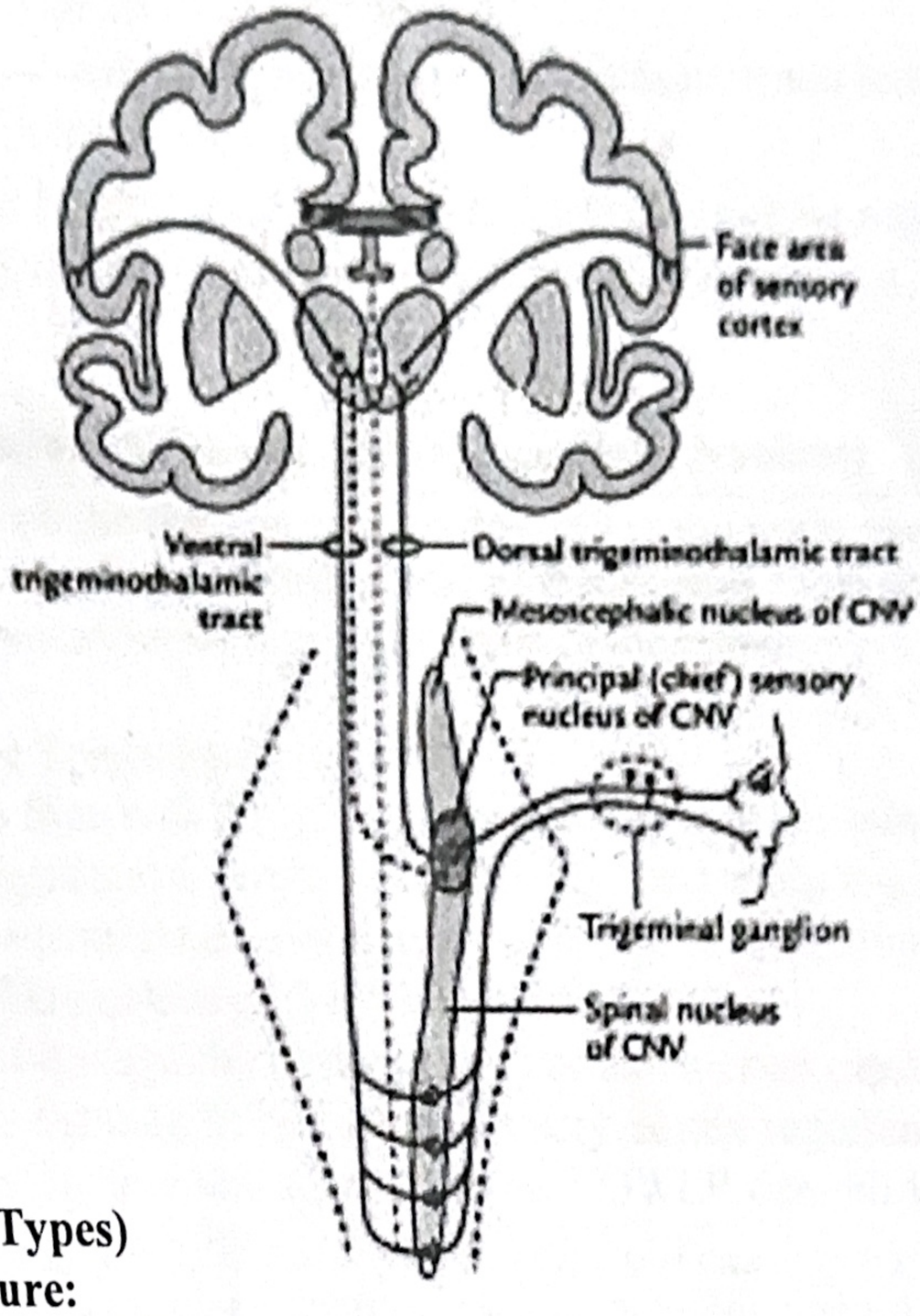
4. PATHWAYS FOR GENERAL SENSATIONS FROM THE HEAD

The head is served by the Trigeminal Nerve (CN V) — the largest cranial nerve — through its three divisions:

- **Ophthalmic (V1)** — forehead, scalp, nose, upper eyelid, cornea
- **Maxillary (V2)** — midface, cheek, upper lip, upper teeth, palate
- **Mandibular (V3)** — lower face, jaw, lower teeth, anterior tongue (touch), temporomandibular joint, and motor to muscles of mastication

Small contributions come from CN VII (facial), CN IX (glossopharyngeal), and CN X (vagus) for limited areas of the face and ear.

Key difference from body pathways: In the head, TOUCH and PROPRIOCEPTION share a different central destination from PAIN and TEMPERATURE — this separation in the brainstem is unique to the trigeminal system.



i. First-Order Neurons (Two Types)

For Touch, Pain, and Temperature:

Cell bodies are in the **Trigeminal (Gasserian / Semilunar) Ganglion** — analogous to the dorsal root ganglion for the body. These are pseudo-unipolar neurons. Their peripheral processes form the three divisions of CN V; their central processes form the **main sensory root of the trigeminal nerve**, which enters the pons.

For Proprioception:

Cell bodies are in the **Mesencephalic Nucleus of the Trigeminal Nerve** in the midbrain. This is the **only first-order sensory neuron found INSIDE the CNS** — a unique anatomical feature. It receives proprioceptive signals from the muscles of mastication, the teeth, the periodontal ligaments, the temporomandibular joint, and the extraocular muscles.

ii. Second-Order Neurons and Their Distinct Nuclei

Depending on the modality carried, first-order axons synapse in DIFFERENT brainstem nuclei:

Touch Fibers → Main (Principal) Sensory Nucleus

Touch fibers from the trigeminal ganglion enter the pons and quickly synapse in the **Main (Principal/Chief) Sensory Nucleus**, located in the pons at the level of CN V entry. This nucleus is analogous to the dorsal column nuclei for the body.

Pain and Temperature Fibers → Spinal Trigeminal Nucleus

Pain and temperature fibers enter the pons but DO NOT synapse there immediately. Instead, they descend as the **spinal trigeminal tract** — traveling down through the medulla, all the way to the upper

ervical spinal cord (C2–C4). They synapse on second-order neurons in the **Spinal Trigeminal Nucleus**.
his nucleus has three subnuclei:

- **Pars oralis (most rostral)** — receives touch and some pain from the mouth region
- **Pars interpolaris** — receives signals from the midface
- **Pars caudalis (most caudal — main pain nucleus)** — receives pain and temperature; equivalent to the dorsal horn of the spinal cord

> *The spinal trigeminal nucleus blends into the substantia gelatinosa of the upper cervical spinal cord — which explains why C2 nerve root pain can mimic facial pain (e.g., the 'cervicogenic headache').*

Proprioception Fibers → Main Sensory Nucleus (via Mesencephalic Nucleus)

The central processes of the mesencephalic nucleus neurons descend to the **main sensory nucleus in the pons** to synapse. Many also project directly to the trigeminal motor nucleus to drive the **jaw-jerk reflex** — a monosynaptic stretch reflex of the masseter muscle, testable in the clinic.

. Third-Order Neurons and the Trigeminal Lemniscus

After synapsing, second-order axons from both the main sensory nucleus and the spinal trigeminal nucleus **cross the midline** and form the **Trigeminal Lemniscus** (also called the ventral trigeminothalamic tract). This tract ascends through the pons and midbrain, running alongside the medial lemniscus, and terminates in the **Ventral Posteromedial (VPM) nucleus of the thalamus**.

Third-order VPM neurons project through the posterior limb of the internal capsule to the **lower one-third of the postcentral gyrus** — the area of the somatosensory cortex representing the face. (Body sensations go to the UPPER two-thirds; face sensations go to the LOWER one-third.)

> *CLINICAL PEARL: Lesions of the TRIGEMINAL NERVE itself cause ipsilateral facial sensory loss. Lesions of the TRIGEMINAL LEMNISCUS cause CONTRALATERAL facial sensory loss. A brainstem lesion affecting both the trigeminal nucleus AND the lateral spinothalamic tract (e.g., Wallenberg/lateral medullary syndrome) causes ipsilateral facial loss + CONTRALATERAL body pain/temperature loss — the classic 'crossed sensory deficit'.*

5. SUMMARY COMPARISON TABLE

The table below consolidates all pathways into a single, easy-to-compare reference:

PATHWAY	MODALITY	1st ORDER (Location)	DECUSSATION LEVEL	2nd ORDER TRACT	CORTEX TARGET
Dorsal Column–Medial Lemniscus	Fine touch, vibration, conscious proprioception, 2-point discrimination	Dorsal root ganglion	Medulla (lemniscal decussation)	Medial lemniscus (Gracile & Cuneate nn.)	Areas 3, 1, 2 (postcentral gyrus)
Anterior Spinothala-mic Tract	Crude touch, pressure	Dorsal root ganglion	Spinal cord (ventral white commissure)	Anterior spino-thalamic tract → spinal lemniscus	Areas 3, 1, 2 (postcentral gyrus)

Lateral Spinothala-mic Tract	Pain, temperature	Dorsal root ganglion	Spinal cord (ventral white commissure)	Lateral spino-thalamic tract → spinal lemniscus	Areas 3, 1, 2 (postcentral gyrus)
Trigeminal (Head)	All general sensations from head & face	Trigeminal ganglion / Mesen-cephalic nucleus	Pons/Medulla (trigeminal lemniscus)	Trigeminal lemniscus (VTT tract)	Lower 1/3 postcentral gyrus

6. CLINICAL CORRELATES — LESIONS OF SENSORY PATHWAYS

Understanding where a pathway decussates tells you exactly what deficit a lesion will cause and on which side. This is the clinical payoff of understanding the anatomy:

LESION SITE	DEFICIT	SIDE AFFECTED
Posterior column (spinal cord)	Loss of fine touch, vibration, proprioception; positive Romberg sign; sensory ataxia	IPSILATERAL, below lesion
Medial lemniscus (brainstem)	Loss of fine touch, vibration, proprioception	CONTRALATERAL body
Lateral spinothalamic tract	Loss of pain and temperature sensation 1–2 segments below lesion	CONTRALATERAL body
Anterior spinothalamic tract	Loss of crude touch and pressure	CONTRALATERAL body
VPL nucleus (thalamus)	Loss of all contralateral somatic sensation; may develop thalamic pain syndrome (Dejerine–Roussy)	CONTRALATERAL body
Trigeminal nerve (CN V)	Loss of all ipsilateral facial sensation (touch, pain, temp, proprioception)	IPSILATERAL face
Trigeminal lemniscus	Loss of all contralateral facial sensation	CONTRALATERAL face
Brown-Séquard syndrome (hemisection)	Ipsilateral loss: fine touch, vibration, proprioception + motor. Contralateral loss: pain and temperature	Mixed – both sides

Key Principle to Remember:

- **Lesions ABOVE the decussation** (e.g., in the thalamus or cortex) → CONTRALATERAL sensory loss for ALL modalities
- **Lesions BELOW the decussation but WITHIN the CNS** → the deficit depends on WHICH tract is affected and WHERE it crosses:
  - Posterior column lesion below the medulla → IPSILATERAL fine touch/vibration/proprioception loss
  - Spinothalamic lesion in the spinal cord → CONTRALATERAL pain/temperature/crude touch loss (1–2 levels below)

7. QUICK MEMORY AID — Pathway Summary in Plain Language

DCML (Fine Touch, Vibration, Proprioception — BODY)

Receptor → Dorsal root ganglion (1st) → Ascends IPSILATERALLY in posterior column → Synapse in Gracile/Cuneate nuclei in MEDULLA (2nd) → Crosses as internal arcuate fibers (lemniscal decussation) → Medial lemniscus ascends → VPL thalamus (3rd) → Posterior limb internal capsule → Upper ⅔ postcentral gyrus (Areas 3, 1, 2)

Lateral Spinothalamic (Pain & Temperature — BODY)

Receptor → Dorsal root ganglion (1st) → Enters cord → Ascends 1–2 levels in Lissauer's tract → Synapse in Substantia Gelatinosa (2nd) → Crosses IMMEDIATELY via ventral white commissure → Lateral spinothalamic tract → Spinal lemniscus → VPL thalamus (3rd) → Postcentral gyrus

Anterior Spinothalamic (Crude Touch & Pressure — BODY)

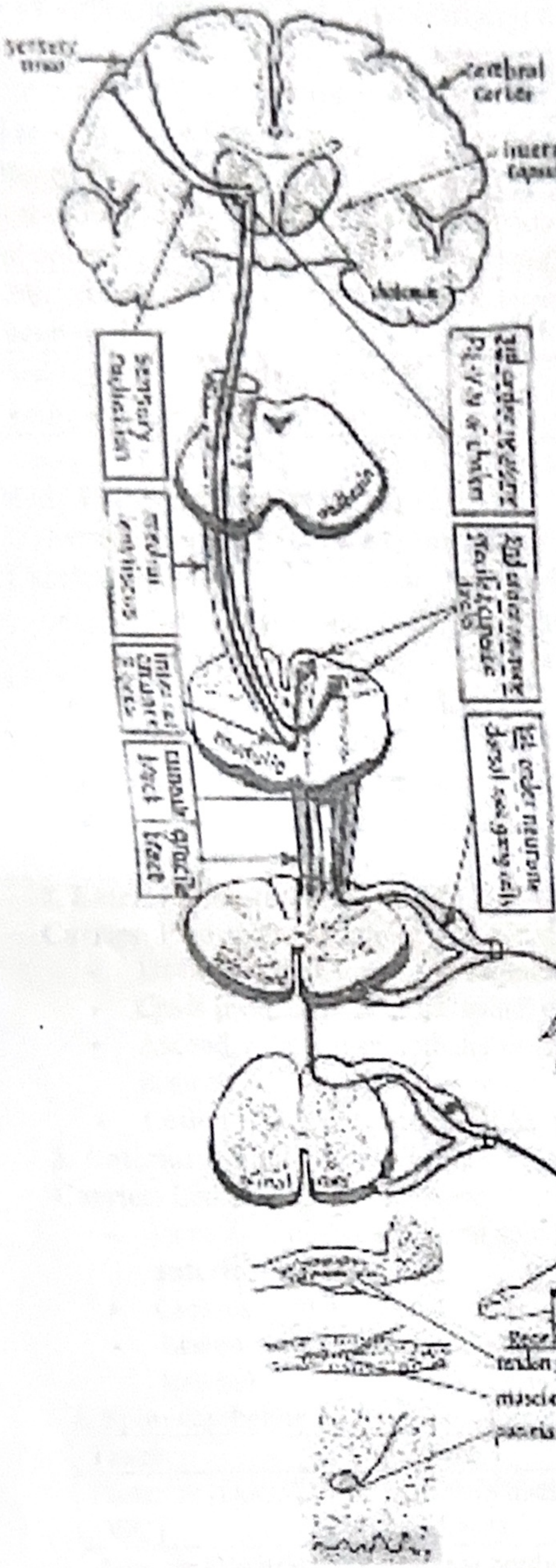
Receptor → Dorsal root ganglion (1st) → Enters cord → Lissauer's tract 1–2 levels → Synapse in Nucleus Proprius (2nd) → Crosses via ventral white commissure → Anterior spinothalamic tract → Spinal lemniscus → VPL thalamus (3rd) → Postcentral gyrus

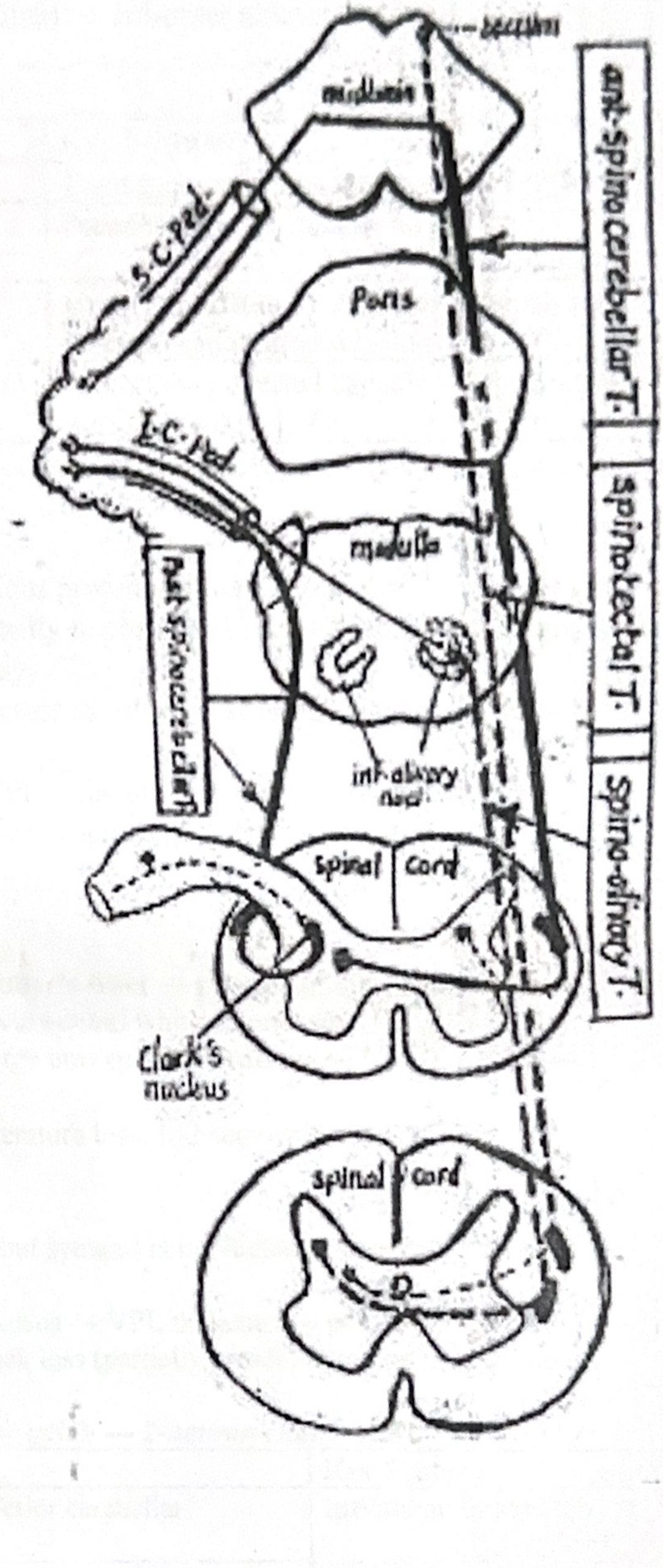
Trigeminal (All General Sensations — HEAD)

Receptors (face/head) → Trigeminal ganglion [or Mesencephalic nucleus for proprioception] (1st) → Enters pons via sensory root:

- **Touch:** → Main Sensory Nucleus (pons, 2nd) → crosses → Trigeminal Lemniscus → VPM thalamus (3rd) → Lower ⅓ postcentral gyrus
- **Pain/Temp:** → Descends as Spinal Trigeminal Tract → Spinal Trigeminal Nucleus (2nd) → crosses → Trigeminal Lemniscus → VPM thalamus (3rd) → Lower ⅓ postcentral gyrus
- **Proprioception:** → Mesencephalic Nucleus (1st, INSIDE CNS) → descends → Main Sensory Nucleus (2nd) → VPM thalamus (3rd) → Lower ⅓ postcentral gyrus

Remember: Every pathway is a three-neuron chain. The key questions are WHERE do the neurons sit, WHERE do they cross, and WHICH SIDE loses function when a lesion occurs.


cerebral cortex
internal capsule
thalamus
sensory radiation
medial lemniscus
internal arcuate fibre
gracile fasc.
spinal cord
receptor:
tendon spindle
muscle spindle
pacinian corpuscle


cerebellum
midbrain
S.C.Ped.
I.C.Ped.
Pons
medulla
Post spinocerebellar T.
inf. olivary nucl.
Spinal Cord
Clark's nucleus
Spino-olivary T.
Spinocerebellar T.
Spinothalamic T.
Spinal Cord

eneral Sensory Pathways — Condensed Master Summary

ore Concept

he nervous system has two functions: receive information (sensory/afferent) and send commands motor/efferent). Sensory pathways carry impulses **upward** from peripheral receptors to the cortex. Every general sensory pathway — regardless of modality — is built from the same **4 elements**: receptor → 1st-order neuron → 2nd-order neuron (crosses midline) → 3rd-order neuron (thalamus) → cortex.

Universal Pathway Architecture

Element	Location	Key Feature
Receptor	Periphery (skin, muscle, joint)	Transduces stimulus into electrical impulse
1st-order neuron	Dorsal root ganglion (body); Trigeminal ganglion (head)	Pseudo-unipolar; outside CNS
2nd-order neuron	CNS (spinal cord/brainstem)	**Crosses midline** — determines which side loses sensation after a lesion
3rd-order neuron	Thalamus (VPL = body; VPM = face)	Projects via internal capsule → postcentral gyrus (Areas 3, 1, 2)

Body Pathways (Below the Head)

1. Dorsal Column–Medial Lemniscus (DCML)

Carries: Fine/discriminative touch, vibration, conscious proprioception, 2-point discrimination

- 1st-order axons enter cord → ascend **ipsilaterally** in posterior column (Fasciculus Gracilis = lower body; Fasciculus Cuneatus = upper body)
- Synapse in Gracile/Cuneate nuclei → axons cross as **internal arcuate fibers** at the **medulla** (lemniscal decussation)
- Form the **Medial Lemniscus** → ascend to VPL thalamus → upper ⅔ postcentral gyrus
- **Lesion = IPSILATERAL** fine touch/vibration/proprioception loss below the level; positive Romberg sign

2. Lateral Spinothalamic Tract

Carries: Pain and temperature

- 1st-order axons travel 1–2 segments in **Lissauer's tract** → synapse in Substantia Gelatinosa
- Cross **immediately** at the spinal cord level via ventral white commissure
- Ascend as lateral spinothalamic tract → merge into **spinal lemniscus** → VPL thalamus → postcentral gyrus
- **Lesion = CONTRALATERAL** pain/temperature loss, 1–2 segments below lesion

3. Anterior Spinothalamic Tract

Carries: Crude touch and pressure

- Same architecture as lateral spinothalamic but synapse is in **Nucleus Proprius**; ascends in the **anterior** funiculus
- Crosses at spinal cord level → spinal lemniscus → VPL thalamus → postcentral gyrus
- **Lesion = CONTRALATERAL** crude touch loss (partially preserved in posterior column lesions)

4. Spinocerebellar Tracts *(Unconscious Proprioception — 2-neuron chains only)*

Tract	Route	Key Feature
Posterior (Dorsal) DSCT	Clarke's nucleus → inferior cerebellar peduncle	Ipsilateral; lower limb
Anterior (Ventral) ASCT	Spinal border cells → crosses **twice**	Ends ipsilateral; double-cross

Cuneocerebellar	Accessory cuneate nucleus → inferior cerebellar peduncle	Upper limb equivalent of DSCT

Head Pathway — Trigeminal System (CN V)

Key Distinction from Body

Touch and proprioception use a **different nucleus** than pain and temperature — they separate within the brainstem.

Modality	1st-order	Synapse (2nd-order)	Crosses
Touch	Trigeminal ganglion	**Main Sensory Nucleus** (pons)	Pons → Trigeminal Lemniscus
Pain/Temperature	Trigeminal ganglion	**Spinal Trigeminal Nucleus** (descends to medulla/C2–C4)	Medulla → Trigeminal Lemniscus
Proprioception	**Mesencephalic nucleus** *(inside CNS — unique!)*	Main Sensory Nucleus	Pons

All three converge → **Trigeminal Lemniscus** → **VPM** thalamus → **lower ⅓** postcentral gyrus

The Single Most Important Clinical Principle

Where a pathway cross determines which side loses sensation after a lesion.

Lesion Location	Modality Lost	Side
Posterior column (cord)	Fine touch, vibration, proprioception	**Ipsilateral**
Lateral spinothalamic (cord)	Pain, temperature	**Contralateral** (1–2 levels below)
Medial/spinal lemniscus, thalamus, cortex	All modalities of contralateral body	**Contralateral**
Trigeminal nerve (CN V)	All facial sensation	**Ipsilateral face**
Trigeminal lemniscus	All facial sensation	**Contralateral face**
Brown-Séquard (cord hemisection)	Fine touch/proprioception/motor (ipsilateral) + Pain/temperature (contralateral)	**Both sides — mixed**
Wallenberg syndrome (lateral medulla)	Ipsilateral face pain/temp + Contralateral body pain/temp	**Crossed deficit**

The One-Line Rule for Every Pathway

Posterior column crosses late (medulla) → ipsilateral cord lesion. Spinothalamic crosses early (cord) → contralateral cord lesion. Everything above the decussation → contralateral loss. Thalamus (VPL/VPM) and above → all modalities, contralateral.